

MICHAEL G. TASSIA, PH.D.

computational biology | data science | statistical genetics

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SUMMARY

- Computational biologist with 10+ years of experience in human genetics, statistical learning, & data science.
- Deep subject matter expertise in somatic mosaicism across human tissues with aging and disease, with specialization in hematology and cancer.
- Proficiency in rapid development of innovative techniques to analyze large and complex genetic data sets, both as a leader and collaborative team-member.
- Productive scientific career underscored by 16 publications, 6 awarded grants, & 18 talks at academic conferences.

EDUCATION & TRAINING

Johns Hopkins University

Postdoctoral Fellow with joint appointments in Biology and Hematology

2022 – Present

Baltimore, MD

Auburn University

Ph.D. in Biological Sciences

2015 – 2021

Auburn, AL

University of Washington

B.S. in Biology (General)

2011 – 2014

Seattle, WA

TECHNICAL SKILLS/EXPERIENCE

Genetics: WGS/WES/RNA-seq data analysis (study design, pre-processing, QC), somatic/germline variant calling (SNVs, SVs, CNVs), phylogenetics, population genetics, QTL mapping/GWAS, differential gene expression

Data science: Regression, classification, inference, prediction, feature selection, non-linear modeling, model validation

Public datasets: 1000 Genomes, All of Us, ClinVar, COSMIC, GENCODE, gnomAD, GTEx, TOPMed, UK Biobank

Programming: R, Bash, Python, Snakemake, Git, high-performance computing and resource scheduling (e.g., slurm)

Bioinformatics software: GATK, bcftools, bedtools, IGV, SLiM, plink, DESeq2, phytools, iqtree

SELECTED AWARDS & HONORS

- **ASH Abstract Achievement Award (2025);** American Society of Hematology
- **Postdoctoral Training Program in Hematology (NIH T32);** Johns Hopkins Division of Hematology & NIH NHLBI
- **ASH Scholar Award (Fellow);** American Society of Hematology

HIGHLIGHTED RESEARCH EXPERIENCE

Modeling evolutionary complexity across hematopoietic progenitors | Johns Hopkins University 2022 – pres.

- Leading an international, multi-institution collaboration to develop and apply novel methods to detect and quantify complex evolutionary histories (e.g., recurrent mutations) among somatic lineages with aging and disease.
- Head developer of Bayesian algorithms tailored to technological advances in single-cell DNA sequencing data.
- Research funded through fellowships from the American Society of Hematology and NIH (T32).

Characterizing the role of telomeres in the evolution of clonal hematopoiesis | Johns Hopkins University 2022 – pres.

- Leveraged expertise in genomics, phylogenetics, and population genetics to reconstruct the evolutionary history of hematopoietic lineages and inform the genetic context of cancer risk.
- Demonstrated that inherited defects in telomere genes affects the evolutionary landscape of hematopoietic stem cells, modulating somatic mutation burden and lifetime risk of cancer.
- Co-first author publication in *NEJM*, establishing a collaboration between labs of co-mentors in the Department of Biology and Department of Oncology at Johns Hopkins School of Medicine.

Studying genetic variation at multiple scales of evolution | Auburn University & Johns Hopkins University 2015 – 2025

- Led and contributed to research studying evolution across diverse timescales, from somatic evolution within human individuals to species divergence across >500 million years.
- *Selected studies include:* Quantification of gene expression variation across human populations; detection of positive selection in complete primate genomes; characterization of innate immunity pathways across the animal tree of life.

PUBLICATIONS & PRESENTATIONS

SELECTED PUBLICATIONS

- Yoo D, Rhie A, Hebbar P, ... **Tassia MG** [86 of 108] ..., *et al.* (2025). Complete sequencing of ape genomes. *Nature*, 461: 401–418. DOI: 10.1038/s41586-025-08816-3
- Makova KD, Pickett BD, Harris RS, ... **Tassia MG** [67 of 88] ..., *et al.* (2024). The complete sequence and comparative analysis of ape sex chromosomes. *Nature* 630: 401–411. DOI: 10.1038/s41586-024-07473-2
- Taylor DJ, Chhetri SB, **Tassia MG**, *et al.* (2024). Sources of gene expression variation in a globally diverse human cohort. *Nature*, 632: 122–130. DOI: 10.1038/s41586-024-07708-2
- DeBoy EA*, **Tassia MG***, Schratz KE, *et al.* (2023). Familial Clonal Hematopoiesis in a Long Telomere Syndrome. *New England Journal of Medicine*, 388: 2422–2433. DOI: 10.1056/NEJMoa2300503. **equal contribution*
- Tassia MG**, Hallowell HA, Waits DS, *et al.* (2023). Induced immune reaction in the acorn worm, *Saccoglossus kowalevskii*, informs the evolution of antiviral immunity. *Molecular Biology & Evolution*, 40. DOI: 10.1093/molbev/msad097
- Tassia MG**, Whelan NV, Halanych KM. (2017). Toll-like receptor pathway evolution in deuterostomes *Proc. Natl. Acad. Sci. U.S.A.*, 114 (27): 7055–7060. DOI: 10.1073/pnas.1617722114

SELECTED CONFERENCE TALKS

- 2025 American Society of Hematology Annual Meeting, Orlando, FL (*Abstract Achievement Award*)
- 2024 Cold Spring Harbor Laboratory: Biology of Genomes, Cold Spring Harbor, NY
- 2023 Society of Molecular Biology & Evolution, Tempe, AZ
- 2023 Society of Integrative & Comparative Biology, Austin, TX (*invited speaker*)
- 2022 Evolution, Cleveland, OH (*virtual attendance*)